Problem 8: That Was an Earthquake? I Barely Felt it
11 Points

Problem ID: `shock`
Rank: 3

Introduction

There is a grid of city blocks. Every block has one seismometer, which remembers the shock level it felt from **only the most recent** earthquake it detected. After a series of earthquakes, what does each seismometer say?

Problem Statement

You are given a city with $\mathbf{N}$ rows and $\mathbf{M}$ columns of blocks. The top left block has coordinates $(0, 0)$ and the bottom right block has coordinates $(\mathbf{N} - 1, \mathbf{M} - 1)$, where the first coordinate is the row and the second coordinate is the column. Every block has exactly one seismometer, initially saying `0`

Then, $\mathbf{Q}$ earthquakes occur. The ith earthquake has an epicenter at coordinates $(\mathbf{R}_i, \mathbf{C}_i)$ and an integer power $\mathbf{P}_i$. For the ith earthquake, the block at row r and column c feels a shock level of:

$$\mathbf{P}_i^2 - \left((r - \mathbf{R}_i)^2 + (c - \mathbf{C}_i)^2\right) + 1$$

In other words, the shock level is the square of the power, minus the square of the block's distance from the epicenter, plus one.

A seismometer updates its reading **only when its corresponding block feels a positive shock level**. Otherwise, the seismometer doesn't realize there was an earthquake at all and just keeps its previous reading. When the seismometer feels a positive shock level, it sets its reading to that shock level. It can be shown that a seismometer will update its reading if and only if its Euclidean distance from the earthquake epicenter is at most $\mathbf{P_i}$.

Display the reading of every seismometer after all earthquakes occur.

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Input Format

The first line of the input contains an integer $\mathbf{T}$ denoting the number of test cases that follow. For each test case:

- The first line contains three space-separated integers $\mathbf{N}$ $\mathbf{M}$ $\mathbf{Q}$ where:
  - $\mathbf{N}$ denotes the number of rows of blocks.
  - $\mathbf{M}$ denotes the number of columns of blocks.
  - $\mathbf{Q}$ denotes the number of earthquakes.
- The next $\mathbf{Q}$ lines contain three space-separated integers $\mathbf{R}_i$ $\mathbf{C}_i$ $\mathbf{P}_i$ and represent an earthquake, given in the order they occur:
  - $\mathbf{R}_i$ denotes the row of the epicenter of the ith earthquake.
  - $\mathbf{C}_i$ denotes the column of the epicenter of the ith earthquake.
  - $\mathbf{P}_i$ denotes the power of the ith earthquake.

Output Format

For each test case, output $\mathbf{N}$ lines containing $\mathbf{M}$ space-separated integers each, denoting the reading of each block's seismometer after all earthquakes have occurred.

Constraints

$1 \leq \mathbf{T} \leq 10^4$
$1 \leq \mathbf{N} \cdot \mathbf{M} \leq 10^6$
$1 \leq \mathbf{Q} \leq 2 \cdot 10^4$
$0 \leq \mathbf{P}_i$
$\mathbf{P}_i \leq \mathbf{R}_i \leq \mathbf{N} - 1 - \mathbf{P}_i$
$\mathbf{P}_i \leq \mathbf{C}_i \leq \mathbf{M} - 1 - \mathbf{P}_i$
It is guaranteed that the sum of $\mathbf{N} \cdot \mathbf{M}$ across all test cases does not exceed 10^6.
It is guaranteed that the sum of $\mathbf{Q}$ across all test cases does not exceed $2 \cdot 10^4$.

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Sample Test Cases

Sample Input Download

```
3
5 6 1
2 3 2
4 3 3
1 1 1
2 1 1
2 1 0
9 9 2
4 4 4
4 3 0
```

Sample Output Download

```
0 0 0 1 0 0
0 0 3 4 3 0
0 1 4 5 4 1
0 0 3 4 3 0
0 0 0 1 0 0
0 1 0
1 1 1
1 1 1
0 1 0
0 0 0 0 1 0 0 0 0
0 0 4 7 8 7 4 0 0
0 4 9 12 13 12 9 4 0
0 7 12 15 16 15 12 7 0
1 8 13 1 17 16 13 8 1
0 7 12 15 16 15 12 7 0
0 4 9 12 13 12 9 4 0
0 0 4 7 8 7 4 0 0
0 0 0 0 1 0 0 0 0
```

Sample Explanations

For the first test case, a single earthquake of power 2 occurs at row 2, column 3. The block right at the epicenter has a shock level of $2^2 - 0 + 1 = 5$, and the block at row 1, column 2 (squared distance 2) has a shock level of $4 - 2 + 1 = 3$. The block at row 1, column 1 (squared distance 5) has a shock level of 0, which is not positive, so that block still reads 0

For the second test case, the first earthquake sets the epicenter at row 1, column 1 to 2 and its four neighbors to 1. The second earthquake, centered at row 2, column 1, sets that block to 2 and its four neighbors to 1, which replaces the 2 at row 1, column 1. But then the third earthquake, with power 0, happens in the same place, replacing its 2 with 1

For the third test case, the first earthquake gives the epicenter at row 4, column 4 a shock level of $4^2 - 0 + 1 = 17$. The second earthquake has power 0, so only the block at row 4, column 3 detects it. That block's reading of 16 is replaced by $0 - 0 + 1 = 1$

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